J. C. BOSE UNIVERSITY OF SCIENCE & TECHNOLOGY, YMCA, FARIDABAD

DEPARTMENT OF COMPUTER ENGINEERING

Progress Report for B.Tech. Capstone Project

2nd Progress Report (Apr. 2026)

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2. Name of Mentor : Dr. Umesh
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5. Title of Project : Pneumonia Disease Detection Using Machine Learning: RespiraCheck

Progress Summary

In the second phase of the RespiraCheck project, the trained DenseNet121 model was wrapped in a production-ready Flask REST API and connected to a fully functional React frontend. A Google Gemini 2.5 Flash AI chat assistant was integrated via the /chat endpoint, enabling patients to interact with their diagnosis result in plain language. The complete full-stack application — covering image upload, real-time prediction with Grad-CAM++ heatmap overlay, and multi-turn AI chat — is functional end-to-end and committed to the project GitHub repository. Concurrently, a recent Nature Scientific Reports paper (Xu & Zhang, 2026) proposing a Double SGAN augmentation framework and ResNet18-SA classifier has been studied to identify further improvements to model performance.

Tasks Completed

- Built the Flask REST API (Flask 3.1.0, Flask-CORS 5.0.1) with three endpoints: /predict (image upload → DenseNet121 inference → Grad-CAM++ heatmap → JSON), /chat (Gemini multi-turn assistant), and /health (liveness probe). Best model loaded at startup from best_model_path.txt; secrets managed via python-dotenv.
- Developed the React (JavaScript/TypeScript) frontend providing an intuitive chest X-ray upload interface. Prediction label, confidence score, and Grad-CAM++ heatmap overlay are displayed side-by-side; a chat window connects to /chat for real-time interaction — no specialist software required, suitable for remote or telemedicine settings.
- Integrated the Google Gemini 2.5 Flash AI chat assistant (google-genai SDK v1.68.0). The assistant supports multi-turn conversation, explains the diagnosis in plain language, educates users on pneumonia symptoms and prevention, and consistently directs users to consult a qualified doctor. System prompt was carefully engineered for safe, non-diagnostic responses.
- Completed end-to-end integration and testing: verified the full pipeline (upload → inference → Grad-CAM++ → JSON → React render), tested /chat with representative patient queries, and confirmed /health liveness responses.
- Full codebase — training notebooks, Flask backend, and React frontend — pushed to GitHub: https://github.com/tanishqsh2k7-source/RespiraCheck.

Tasks in Progress

- Investigating **GAN-based synthetic data augmentation** inspired by Xu & Zhang (Scientific Reports, 2026), which proposed a Double SGAN model using spectral normalisation, self-attention, and hinge loss to generate balanced minority-class X-rays. Preliminary experiments suggest this approach could reduce the false-negative rate beyond what class weights alone achieve, targeting pneumonia recall above 93%.
- Evaluating **hinge loss as a replacement for binary cross-entropy** during fine-tuning. The Nature paper shows hinge loss prevents gradient saturation near decision boundaries and maintains large gradient signals for the minority class (NORMAL), which could directly address the precision-recall trade-off observed in RespiraCheck's current DenseNet121 evaluation.
- Researching **multi-class classification extension** using the NIH ChestX-ray14 dataset to distinguish bacterial pneumonia, viral pneumonia, COVID-19, and tuberculosis — identified as the primary future-work direction.
- Preparing the **final project report and presentation** for end-of-semester capstone submission; finalising performance comparison tables and Grad-CAM++ visualisation figures for inclusion in the report.

Date: ……………… **(Signature of Student)**

Report of the Mentor

(to be filled by mentor)

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Signature of Mentor